

Classification

Binary Tree

Tree Construction

age	income	student	credit rating	buys computer
<=30	high	no	fair	no
<=30	high	no	excellent	no
31...40	high	no	fair	yes
>40	medium	no	fair	yes
>40	low	yes	fair	yes
>40	low	yes	excellent	no
31...40	low	yes	excellent	yes
<=30	medium	no	fair	no
<=30	low	yes	fair	yes
>40	medium	yes	fair	yes
<=30	medium	yes	excellent	yes
31...40	medium	no	excellent	yes
31...40	high	yes	fair	yes
>40	medium	no	excellent	no

$$Entropy(t) = - \sum_j p(j|t) \log_2 p(j|t)$$

$$Gain_{split} = Entropy(p) - \left(\sum_{i=1}^k \frac{n_i}{n} Entropy(i) \right)$$

$$|buys\ computer| = 14 \rightarrow [|Yes| = 9, |No| = 5]$$

$$Entropy(t) = -\frac{9}{14} \log_2 \left(\frac{9}{14} \right) - \frac{5}{14} \log_2 \left(\frac{5}{14} \right) = 0.940$$

age	Yes	No	n_i
<=30	2	3	5
31...40	4	0	4
>40	3	2	5

$$Entropy(age \leq 30) = -\frac{2}{5} \log_2 \left(\frac{2}{5} \right) - \frac{3}{5} \log_2 \left(\frac{3}{5} \right) = 0.971$$

$$Entropy(age\ 31...40) = 0$$

$$Entropy(age > 40) = Entropy(age \leq 30) = 0.971$$

$$Gain(age) = 0.940 - \left[\frac{5}{14} (0.971) + 0 + \frac{5}{14} (0.971) \right] = 0.246$$

income	Yes	No	n_i
low	3	1	4
medium	4	2	6
high	2	2	4

$$Entropy(income\ low) = -\frac{3}{4} \log_2 \left(\frac{3}{4} \right) - \frac{1}{4} \log_2 \left(\frac{1}{4} \right) = 0.811$$

$$Entropy(income\ medium) = -\frac{4}{6} \log_2 \left(\frac{4}{6} \right) - \frac{2}{6} \log_2 \left(\frac{2}{6} \right) = 0.918$$

$$Entropy(income\ high) = 1$$

$$Gain(income) = 0.940 - \left[\frac{4}{14} (0.811) + \frac{6}{14} (0.918) + \frac{4}{14} (1) \right] = 0.029$$

student	Yes	No	n_i
Yes	6	1	7
No	3	4	7

$$Entropy(student\ Yes) = -\frac{6}{7} \log_2 \left(\frac{6}{7} \right) - \frac{1}{7} \log_2 \left(\frac{1}{7} \right) = 0.592$$

$$Entropy(student\ No) = -\frac{3}{7} \log_2 \left(\frac{3}{7} \right) - \frac{4}{7} \log_2 \left(\frac{4}{7} \right) = 0.985$$

$$Gain(student) = 0.940 - \left[\frac{7}{14} (0.592) + \frac{7}{14} (0.985) \right] = 0.152$$

credit rating	Yes	No	n_i
fair	6	2	8
excellent	3	3	6

$$Entropy(c.r.\ fair) = -\frac{6}{8} \log_2 \left(\frac{6}{8} \right) - \frac{2}{8} \log_2 \left(\frac{2}{8} \right) = 0.811$$

$$Entropy(c.r.\ excel) = 1$$

$$Gain(c.r.) = 0.940 - \left[\frac{8}{14} (0.811) + \frac{6}{14} (1) \right] = 0.048$$

The Information Gain of age is the highest, we choose age as a root node

age decode	
0.5 <	0 : 31...40
1.5 <	1 : <= 30
2	2 : >40

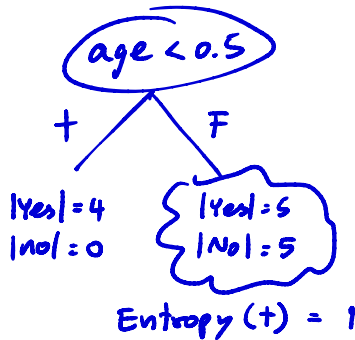
age < 0.5	Yes	No	n _i	ent _i
31...40	4	0	4	0
<= 30	5	5	10	1
>40				

$$\text{Gain}(\text{age} < 0.5) = 0.940 - \left[0 + \frac{10}{14} (1) \right] = 0.226$$

age < 1.5	Yes	No	n _i	ent _i
31...40	6	3	9	0.918
<= 30	3	2	5	0.971
>40				

$$\text{Gain}(\text{age} < 1.5) = 0.940 - \left[\frac{9}{14} (0.918) + \frac{5}{14} (0.971) \right] = 0.003$$

∴ Gain(age < 0.5) is the highest so we choose age < 0.5 as the root node



$$\text{Entropy}(T) = 1$$

Next we consider which feature should be node of the right child

Income	Yes	No	n _i	ent _i
low	2	1	3	0.918
med	3	2	5	0.971
high	0	2	2	0

$$\text{Gain}(\text{income}) = 1 - \left(\frac{3}{10} (0.918) + \frac{5}{10} (0.971) \right) = 0.239$$

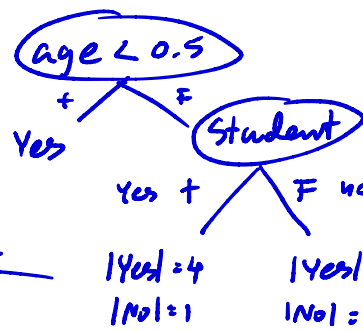
∴ student has the highest gain

student	Yes	No	n _i	ent _i
yes	4	1	5	0.722
no	1	4	5	0.722

$$\text{Gain}(\text{stu}) = 1 - \left(\frac{5}{10} (0.722) + \frac{5}{10} (0.722) \right) = 0.278$$

cr	Yes	No	n _i	ent _i
fair	4	2	6	0.918
excell	1	3	4	0.811

$$\text{Gain}(\text{cr}) = 1 - \left(\frac{6}{10} (0.918) + \frac{4}{10} (0.811) \right) = 0.125$$



$$\text{ent}(F) = 0.722$$

income	Yes	No	n _i	ent _i
low	2	1	3	0.918
medium	2	0	2	0

$$\text{gain}(\text{income}) = 0.722 - \left[\frac{3}{5} (0.918) \right] = 0.171$$

income	Yes	No	n _i	ent _i
high	0	2	2	0
med	1	2	3	0.918

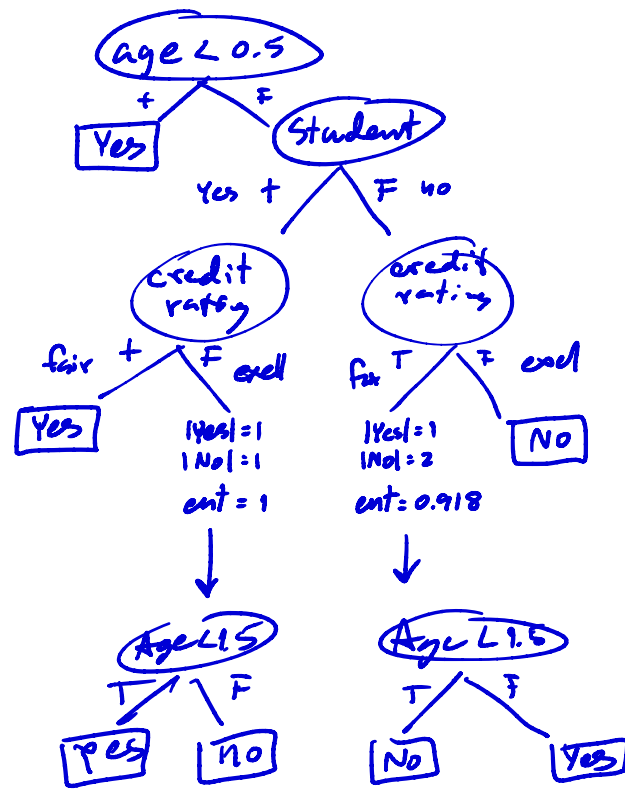
$$\text{Gain}(\text{income}) = 0.171$$

cr	Yes	No	n _i	ent _i
fair	3	0	3	0
excell	1	1	2	1

$$\text{Gain}(\text{cr}) = 0.722 - \left[\frac{3}{5} (1) \right] = 0.322$$

cr	Yes	No	n _i	ent _i
fair	1	2	3	0.918
excell	0	2	2	0

$$\text{Gain}(\text{income}) = 0.171$$



age	income	student	credit rating	buys computer
<=30	high	no	fair	no
<=30	high	no	excellent	no
31...40	high	no	fair	yes
>40	medium	no	fair	yes
>40	low	yes	fair	yes
>40	low	yes	excellent	no
31...40	low	yes	excellent	yes
<=30	medium	no	fair	no
<=30	low	yes	fair	yes
>40	medium	yes	fair	yes
<=30	medium	yes	excellent	yes
31...40	medium	no	excellent	yes
31...40	high	yes	fair	yes
>40	medium	no	excellent	no

age	Yes	No	n_i	ent_i
<=30	2	3	5	0.971
31...40	4	0	4	0
>40	3	2	5	0.971

$$Gain(age) = 0.94 - \left[\frac{5}{14}(0.971) + \frac{4}{14}(0) + \frac{5}{14}(0.971) \right]$$

$$= 0.246$$

income	Yes	No	n_i	ent_i
low	3	1	4	0.811
med	4	2	6	0.918
high	2	2	4	1

$$Gain(income) = 0.94 - \left[\frac{4}{14}(0.811) + \frac{6}{14}(0.918) + \frac{4}{14}(1) \right]$$

$$= 0.029$$

student	Yes	No	n_i	ent_i
Yes	6	1	7	0.592
No	3	4	7	0.985

$$Gain(student) = 0.94 - \left[\frac{7}{14}(0.592) + \frac{7}{14}(0.985) \right]$$

$$= 0.1515$$

credit rating	Yes	No	n_i	ent_i
fair	6	2	8	0.811
excellent	3	3	6	1

$$Gain(credit rating) = 0.94 - \left[\frac{8}{14}(0.811) + \frac{6}{14}(1) \right]$$

$$= 0.048$$

$\therefore$ age has the highest gain, so age is the root node

Left child

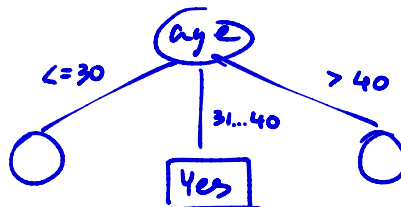
Consider 'student'

student	Yes	No	n_i	ent_i
Yes	2	0	2	0
No	0	3	3	0

$$\therefore \sum_{i=1}^2 \frac{n_i}{n} \text{Entropy}(i) = 0$$

$\therefore$ Gain is maximum

Left child is 'student' is the node



Right child

Consider 'credit rating'

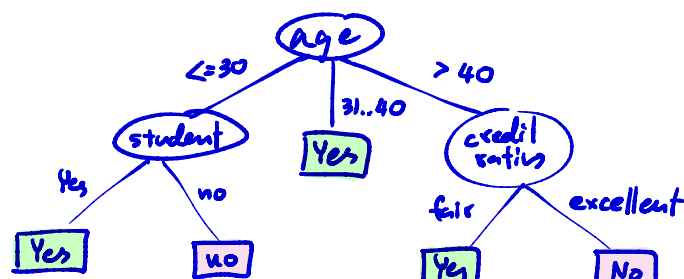
CR	Yes	No	n_i	ent_i
fair	3	0	3	0
excellent	0	2	2	0

$$\therefore \sum_{i=1}^2 \frac{n_i}{n} \text{Entropy}(i) = 0$$

$\therefore$ Gain is maximum

Right child is 'credit rating' is the node

Result



Parameter Tuning

		Predicted	
		Spam	Non-spam
Actual	Spam	600 TP	300 FN
	Non-spam	100 FP	9000 TN

$$\text{Accuracy} = \frac{TP+TN}{TP+FN+FP+TN} = \frac{9600}{10000} = 0.96$$

$$\text{Recall} = \frac{TP}{TP+FN} = \frac{600}{900} = 0.67$$

$$\text{Precision} = \frac{TP}{TP+FP} = \frac{600}{700} = \frac{6}{7}$$

$$F1 = \frac{2 \cdot PR}{P+R}$$

	Actual Cancer = Yes	Actual Cancer = No
Predicted Cancer = Yes	True Positive 57	False Positive 14
Predicted Cancer = No	False Negative 23	True Negative 171

$$\text{Accuracy} = \frac{TP+TN}{TP+FP+FN+TN} = \frac{57+171}{57+14+23+171}$$

$$\text{Recall} = \frac{TP}{TP+FN} = \frac{57}{57+23}$$

$$\text{Precision} = \frac{TP}{TP+FP} = \frac{57}{57+14}$$

$$F1 = \frac{2PR}{P+R}$$

Linear Regression

x	y_true	y_pred
1	3	2.5
2	5	5.4
3	2	2.2
4	8	7.8

R square

$$SSE = \sum_{i=1}^4 (y_i - \hat{y}_i)^2 = 0.25 + 0.16 + 0.04 + 0.04 = 0.49$$

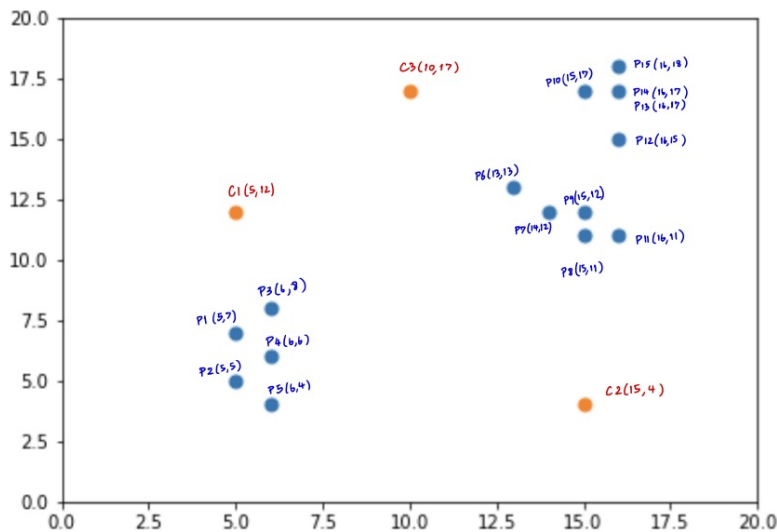
$$\hat{y} = \frac{3+5+2+8}{4} = 4.5$$

$$SST = \sum_{i=1}^4 (y_i - \bar{y})^2 = 1.5^2 + 0.5^2 + 2.5^2 + 3.5^2 = 2.25 + 0.25 + 6.25 + 12.25 = 21$$

$$R^2 = 1 - \frac{SSE}{SST} = 1 - \frac{0.49}{21} = 0.977$$

Clustering

Given data points and initial centroids (in orange) below, use the k-means clustering algorithm discussed in the lecture to compute new centroids after 3 iterations. Show your work in each step.



Let $k=3$

Data points:

[5, 7], [5, 5], [6, 6], [6, 4], [6, 4], [13, 13], [14, 12], [15, 11], [15, 11], [15, 12], [15, 17], [16, 17], [16, 17], [16, 18]

Initial centroids:

[5, 12], [15, 4], [10, 17]

Iteration 1

For P7

$$C2, P7 \Rightarrow \sqrt{(15-14)^2 + (4-12)^2} = 8.062$$

$$C3, P7 \Rightarrow \sqrt{(10-14)^2 + (17-12)^2} = 6.403$$

For P8

$$C2, P8 \Rightarrow \sqrt{(15-15)^2 + (4-11)^2} = 7$$

$$C3, P8 \Rightarrow \sqrt{(10-15)^2 + (17-11)^2} = 7.810$$

For P9

$$C2, P9 \Rightarrow \sqrt{(15-15)^2 + (4-12)^2} = 8$$

$$C3, P9 \Rightarrow \sqrt{(10-15)^2 + (17-12)^2} = 7.071$$

For P11

$$C2, P11 \Rightarrow \sqrt{(15-16)^2 + (4-11)^2} = 7.071$$

$$C3, P11 \Rightarrow \sqrt{(10-16)^2 + (17-11)^2} = 8.425$$

Cluster 1 $\Rightarrow$ P1, P2, P3, P4, P5

Cluster 2 $\Rightarrow$ P8, P11

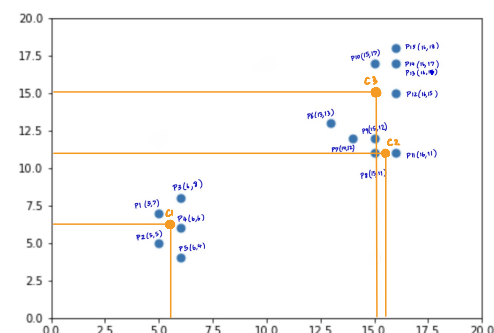
Cluster 3 $\Rightarrow$ P6, P7, P9, P10, P12, P13, P14, P15

Find new centroid

$$\text{centroid 1} = \left(\frac{5+5+6+6+6}{5}, \frac{7+5+8+6+4}{5} \right) = (5.6, 6)$$

$$\text{centroid 2} = \left(\frac{15+16}{2}, \frac{11+11}{2} \right) = (15.5, 11)$$

$$\text{centroid 3} = (15.125, 15.125)$$



Iteration 2

For P6

$$C_2, P_6 \Rightarrow \sqrt{(15.5-13)^2 + (11-13)^2} = 3.202$$

$$C_3, P_6 \Rightarrow \sqrt{(15.115-13)^2 + (11.125-13)^2} = 3.005$$

Cluster 1 $\Rightarrow P_1, P_2, P_3, P_4, P_5$

Cluster 2 $\Rightarrow P_7, P_8, P_9, P_{11}$

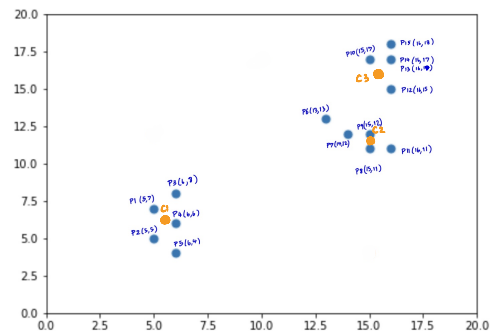
Cluster 3 $\Rightarrow P_6, P_{10}, P_{12}, P_{13}, P_{14}, P_{15}$

compute new centroid

centroid₁ $\Rightarrow (5.6, 6)$

centroid₂ $\Rightarrow (15, 11.5)$

centroid₃ $\Rightarrow (15.33, 16.17)$



Iteration 3

For P6

$$C_2, P_6 \Rightarrow \sqrt{(15-13)^2 + (11.5-13)^2} = 2.5$$

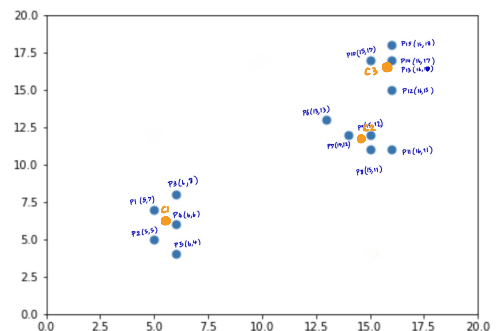
$$C_3, P_6 \Rightarrow \sqrt{(15.33-13)^2 + (16.17-13)^2} = 3.934$$

compute new centroid

centroid₁ : (5.6, 6)

centroid₂ : (14.6, 11.8)

centroid₃ : (15.8, 16.8)



WCSS

$$\text{Cluster 1} : 1.36 + 1.36 + 4.16 + 0.16 + 4.16 = 11.17$$

$$\text{Cluster 2} : 4 + 0.4 + 0.2 + 0.8 + 2.6 = 8$$

$$\text{Cluster 3} : 0.68 + 0.08 + 0.08 + 3.38 + 1.48 = 5.6$$

} 24.77

$$\text{WCSS} = 24.77$$

WCSS

Let $K=2$

Cluster 1 : A, B, C

Cluster 2 : D, E, F

A	(1, 2)
B	(1, 4)
C	(1, 0)
D	(10, 2)
E	(10, 4)
F	(10, 0)

Centroid₁ : (1, 2)

Centroid₂ : (10, 2)

Square distance from centroid

Cluster 1 $0 + 4 + 4 = 8$

Cluster 2 $0 + 4 + 4 = 8$

WCSS_{total} = 16

Agglomerative

P1	(0.4, 0.53)
P2	(0.22, 0.38)
P3	(0.35, 0.32)
P4	(0.26, 0.19)
P5	(0.08, 0.41)
P6	(0.45, 0.30)

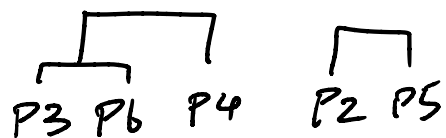
	P1	P2	P3	P4	P5
P1					
P2	0.23				
P3	0.22	0.14			
P4	0.37	0.19	0.16		
P5	0.34	0.14	0.28	0.28	
P6	0.24	0.24	0.1	0.22	0.39

	P1	P2	P3 P6	P4	P5
P1					
P2	0.23				
P3 P6	0.24	0.24			
P4	0.37	0.19	0.22		
P5	0.34	0.14	0.39	0.28	

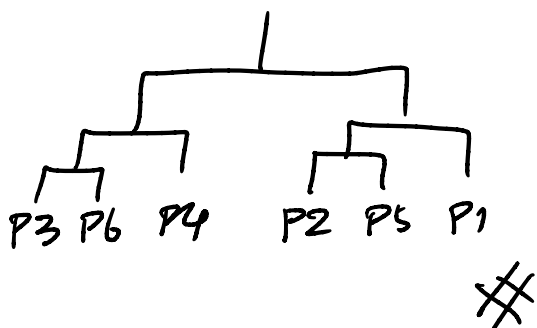
P3 P6



	P1	P2 P5	P3 P6	P4
P1				
P2 P5	0.34			
P3 P6	0.24	0.39		
P4	0.37	0.28	0.22	



	P1	P2 P5	P3 P6 P4
P1			
P2 P5	0.34		
P3 P6 P4	0.37	0.39	

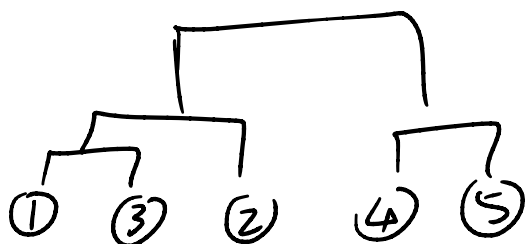
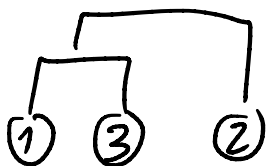


	P1 P2 P5	P3 P6 P4
P1 P2 P5		
P3 P6 P4		

(x_2, y_2)

ลำดับ (x,y)	1 (32,54)	2 (64,62)	3 (11,30)	4 (102,120)	5 (70,105)
1 (32,54)	0	32.98	31.89	96.21	63.60
2 (64,62)	32.98	0	61.91	69.34	43.42
3 (11,30)	31.89	61.91	0	127.99	95.43
4 (102,120)	96.21	69.34	127.99	0	35.34
5 (70,105)	63.60	43.42	95.43	35.34	0

(x_1, y_1)



	1,3	2	4	5
1,3				
2	32.98			
4	46.21	69.34		
5	63.60	43.42	35.34	

	1,3 2	4	5
1,3 2			
4	63.34		
5	43.42	35.34	